# C17 Memory Safety, Pointer Arithmetic & Sanitizers

## KEY SYNTAX & IDIOMS

**Core Syntax Declaration**
```
function execute() { return true; }
```
Standard function signature

**Assertion Verification**
```
assert(result === expected);
```
Validates post-conditions

**Error Handling**
```
try { ... } catch (e) { ... }
```
Graceful failure recovery

**Scoped Constants**
```
const CONFIG = { mode: 'sandbox' };
```
Immutable reference declaration

## COMMON PITFALLS & FIXES

**Pitfall:** Missing type assertions
**Rule:** Enforce strict compiler lint checks and compiler warnings

**Pitfall:** Uncontained side effects
**Rule:** Run untrusted student snippets in isolated container sandboxes

**Pitfall:** Unbounded recursion
**Rule:** Set configurable 20s timeouts to contain infinite loops

## TINY VERIFIED EXAMPLE: VERIFIED EXECUTION

**SOURCE CODE:**
```
console.log('Verified Course Material Studio');
```

**REAL CAPTURED OUTPUT:**
```
Verified Course Material Studio
```